

Machine Learning-Enabled Localization in 5G using LIDAR and RSS Data

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Abstract—Demand for localization has been growing due to the increase in location-based services and high bandwidth applications requiring precise localization of users to improve resource management and beam forming. Outdoor localization has been traditionally done through Global Positioning System (GPS), however its performance degrades in urban settings due to obstruction and multi-path effects, creating the need for better localization techniques. We propose a technique using a cascaded approach composed of image classification and regional convolutional neural networks (CNNs) using LIDAR and received signal strength (RSS) data to predict the location of moving vehicles outdoors. We use simulated data in the mmWave band that takes place in the neighbourhood of Rosslyn in Arlington, Virginia. Our results show an improvement in localization accuracy as a result of the hierarchical architecture, with a mean absolute error (MAE) of 6.55m for the proposed technique in comparison to a MAE of 9.82m using one CNN.

Index Terms—localization.

I. INTRODUCTION

Accurate localization of users, devices and vehicles has been an active research area with outcomes impacting many applications such as marketing, self-driving cars, smart parking and so on. Localization is simply the process of retaining a user's location. Traditionally, outdoor localization is done through GPS. GPS has good accuracy for unobstructed-sky however its performance degrades in urban canyons as a result of severe obstruction and multipath effects. On the other hand, indoor localization can be done using Wi-Fi or Bluetooth signals, using three main methods: Trilateration, fingerprinting and proximity. Trilateration is very similar to GPS; in that it uses three transmitters with known locations to find the receiver [1]. This method uses information such as RSS, time of arrival (ToA), angle of arrival (AoA) or delay. The fingerprinting approach collects information regarding received signals and their locations during the online or training phase and compares RSS in the offline phase to estimate the location [2]. Finally, proximity methods report when devices are in range, and estimate the location of the device based on their closeness to known locations. Although effective, these techniques lack accuracy and rely on the user being connected to Wi-Fi or Bluetooth, or within range to nearby sensors to estimate their location. Moreover, GPS and other satellite signals lack precision or entirely fail indoors [3]. There is clearly a need for accurate localization both indoors and outdoors. In this paper, we focus on outdoor localization however the developed methods are applicable to indoor localization.

Localization through mobile network (LTE, 5G) base stations is a particularly intriguing topic as the applications go beyond navigation. Knowledge of the user's location assists the base station with accurately estimating channel conditions, thus improving resource management and beam forming [4]. As 5G requirements go beyond daily cellphone usage and aim to achieve enhanced mobile broadband, massive machine type communications and ultra-reliable and low latency communications, there is a growing need to advance localization technologies to support high frequency spectrum [5]. The mmWave spectrum has gained significant attention in the upcoming 5G networks due to its high data transmission rates, which will be able to support such applications. However, the mmWave band comes with its own set of challenges. For example, the attenuation properties of the channel increase the effects of blockage as mmWaves cannot travel through solid mediums [6]. High mobility applications also create the need for frequent beam tracking [6]. Moreover, the narrow beams require precise localization methods as errors in location can impact the transmission.

Machine learning has been used in many applications such as speech and image recognition and financial predictions, and within the last few years, has been a hot topic in wireless networks. Recent research has shown the benefits of machine learning when applied to problems such as routing, resource allocation and adaptive modulation techniques [5] [7] [8]. In this paper, we address the above issues by introducing a localization algorithm using a hierarchical regional classification and Convolutional Neural Network (CNN) approach. We use RSS, AoA, AoD and ToA data in conjunction with lidar data collected from a simulated dataset that takes place in a neighbourhood in Rosslyn, Virginia and predict the location of moving vehicles within the provided map. We execute the proposed technique by first dividing the map into 3 different regions that are visually distinguishable through their landmarks. Next, we implement an image classification algorithm using image differencing that takes lidar images as an input to label the user's region based on the predefined boundaries. Finally, we train one CNN for each region of the map using training data that is distinct to that region.

The rest of the paper is organized as follows. In section II we summarize the state of the art and identify the differences with our work. In section III we explain the details of the data set, the generation process and the environment where the localization takes place. The proposed solutions are discussed in section IV and the results of the varying network architectures

are shown in section V. Finally, in Section VI we conclude the paper.

II. RELATED WORK

Localization is a widely researched topic in wireless communications. While traditional approaches explore the use of GPS for outdoor applications; and trilateration, proximity and fingerprinting for indoor applications, machine learning-based localization techniques have been increasingly prominent in recent works. Some of these studies exploit the user data that is available to the base station or access point by establishing a communication link such as RSS, ToA, AoA and AoD. Whereas, other studies utilize sensors and cameras equipped on the user's devices to predict locations such as in [9]. A variety of neural network structures have been researched such as in [10], [11], [12] and [13], particularly Feed-Forward Neural Networks (FFNN), CNNs and long-short term memory (LSTMs) neural networks.

For instance, [10] proposes an approach using time-series RSS readings from WLAN access points and CNNs to predict buildings, floors and location co-ordinates. They take a hierarchical approach in which they first predict the building, then use the building's specific model to predict the floor and finally use the floor's model to predict the location co-ordinates. They create an $N \times T$ RSS feature image using time series RSS readings periodically collected from mobile nodes and use it as an input to the CNN. Their work demonstrates the improvements in localization error as a result of using time-series RSS readings in conjunction with CNNs rather than single vector RSS readings with a feed-forward neural network. While this approach achieves exceptional results, the concept is difficult to implement in high mobility applications, such as vehicular communications. Our proposed technique uses a CNN input feature matrix composed of single-instance RSS measurements along with AoA, AoD and ToA from 25 different antennas.

[9] uses camera images along with CNNs for landmark detection and localization to avert the use of weak or less accurate GPS signals outdoors. They propose two localization methods in which they investigate the object detection capabilities of CNNs and Faster Region-based CNNs (R-CNNs). Their localization technique relies heavily on the images and predicts locations based on the picture's proximity to predefined landmarks. In this paper, we propose a solution that uses image classification to predict predefined regions rather than location co-ordinates. The precise location is then predicted using a CNN that is trained using data from that distinct region. This structure allows us to gain the benefits of the image detection techniques while maintaining a lower localization error.

III. DATASET

We use the Raymobtime-s008 and s009 datasets that are available at [14]. The datasets provide realistic data from a simulated 5G network using millimeter wave (mmWave) and multiple input, multiple output (MIMO). This data is generated

for over a terrain size of 300 m by 900 m that maps onto Rosslyn which is an urban neighborhood in Arlington, VA. In this scenario, the receivers are a combination of moving cars, trucks and buses, each with a dedicated speed. Fig. 1 displays the Rosslyn map, where the blue entities represent the building structures, the red data points represent the positions of the vehicles on the streets and the green point represents the location of the base station. The dataset contains ray-tracing data, lidar data [15] and user position information. The dataset contains the following ray-tracing data: Received power, ToA, elevation AoA, azimuth AoA, elevation AoD, azimuth AoD, gain phase, ray phase and LOS flag. The following user information is also provided: episode ID, scene ID, vehicle array ID, vehicle name, LOS flag, x, y and z co-ordinated.

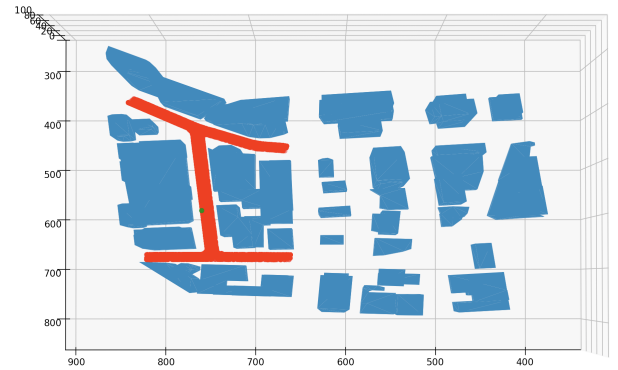


Fig. 1. Plotted map of Rosslyn, Virginia (blue) and data set points (red), base station (green)

Both datasets contain approximately 20000 samples. While extracting the data, samples containing invalid channels were discarded, leaving a remainder of 11194 samples for the s008 dataset and 9638 samples for the s009 dataset. In this scenario, invalid channels are defined as communication channels belonging to vehicles that have left the analysis zone. The datasets are then split into a 60/40 ratio for the training and testing samples respectively. The pre-processing stage consists of replacing "NaNs" or invalid values in the dataset with 0s, min-max normalization, and the concatenation of the features into a single matrix. The matrix, shown in eq.1, is composed of 7 rows and 25 columns, with each row containing a feature, and the columns representing measurements from each antenna. The lidar point cloud representation data was converted into coloured bird's eye view images. Several steps were taken to process the images. First, the image is converted into grayscale to reduce dimensionality. Then, the image is blurred to remove intricate detail and is then converted to black and white. The white space in the image background is cropped. Next, the images are padded again with a white background. This is done to center all of the images in a uniform manner. Finally, the images are scaled down to reduce computational complexity.

$$\begin{bmatrix} \text{AoA_el_Ant0} & \text{AoA_el_Ant1} & \dots & \text{AoA_el_Ant24} \\ \text{AoA_az_Ant0} & \text{AoA_az_Ant1} & \dots & \text{AoA_az_Ant24} \\ \text{AoD_el_Ant0} & \text{AoD_el_Ant1} & \dots & \text{AoD_el_Ant24} \\ \text{AoD_az_Ant0} & \text{AoD_az_Ant1} & \dots & \text{AoD_az_Ant24} \\ \text{RSS_Ant0} & \text{RSS_Ant1} & \dots & \text{RSS_Ant24} \\ \text{gainPhase_Ant0} & \text{gainPhase_Ant1} & \dots & \text{gainPhase_Ant24} \\ \text{ToA_Ant0} & \text{ToA_Ant1} & \dots & \text{ToA_Ant24} \\ \text{RayPhase_Ant0} & \text{RayPhase_Ant1} & \dots & \text{RayPhase_Ant24} \end{bmatrix} \quad (1)$$

IV. PROPOSED MACHINE LEARNING-ENABLED LOCALIZATION TECHNIQUE

This paper proposes a cascaded image classification and deep learning algorithm using lidar images and ray-tracing data to predict user locations. This is a centralized technique in which the vehicles provide the required information to a localization service provider where all the computations take place. To reduce localization error, the approach introduces CNNs that correspond to a predefined region in the map. The regional neural networks are trained using samples belonging to that specific region, rather than utilizing the entire training sample set, which allows the predictor to learn the trends and outcomes distinct to that region. Our algorithm uses RSS-based fingerprinting along with image detection to automatically sort the data points to the correct region. This is done through image differencing, which compares the lidar images provided by the vehicles to a repository of stored images of various map regions. The regional classification algorithm is described in the flowchart in Fig. 2. Fig. 3 displays the 3 regions in the map. All the regions are within the coverage area of the same base station. Our scheme considers two approaches for defining the regions. The first approach has solid defined boundaries, i.e., there is no overlap between neighbouring regions. In the second approach, the boundaries of neighbouring regions are extended by approximately 50m to create an overlap. This is done to reduce the chances of false image classification in regional boundaries.

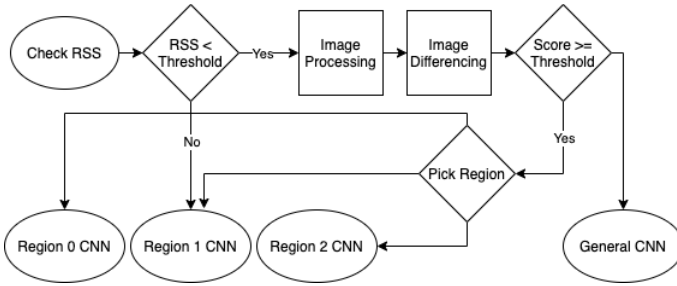


Fig. 2. Flowchart displaying hybrid RSS based fingerprinting and image classification approach.

A. Regional Classification

Regional classification is the first segment of the cascaded localization technique. We propose a hybrid classification approach comprised of RSS based classification and lidar image detection. The RSS values are plotted to analyze the correlation between location and RSS strength. Based on the selected dataset, it is determined that region 1 contains the strongest RSS values due to its proximity to the transmitter. Regions 0 and 2 have weaker RSS values, making them more

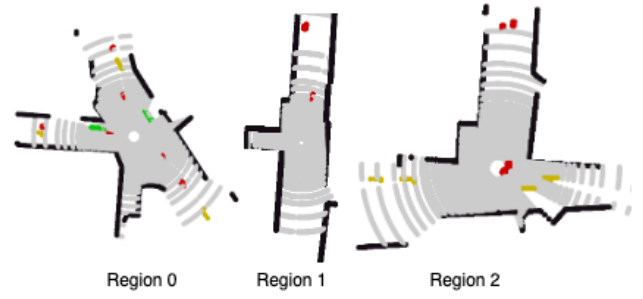


Fig. 3. Sample lidar and training images for the 3 different map regions

difficult to classify using fingerprinting. Hence, they only use the lidar images for regional classification. The hybrid approach reduces the complexity of the algorithm and the chances of image classification errors by only relying on lidar images when needed. In Algorithm 1 we start by checking the RSS value of the sample. If the normalized RSS is greater than or equal to the defined threshold, then the sample belongs to region 1. Otherwise, the lidar images are processed and image differencing is used to compare the lidar image to the repository of images. The image differencing algorithm provides a score which defines the correlation between the two images, the closer the value is to 1, the stronger the correlation. The predicted region is the region of the image with the strongest score. Fig. 3 displays a sample of the training images used for the 3 regions.

Algorithm 1 Localization Algorithm

Data: Raytracing Data, Lidar Images

Result: X, Y, Z Coordinates

initialization **for Every data point do**

check SNR **if** $SNR \geq Threshold$ **then**

Append data to region 1 CNN, Predict Location

else

for Every Training Image do

checkScore(currentImage, TrainingImage)

end

if Image Score $\leq Threshold$ **then**

Append data to general CNN, predict location

else

Region = index(max(imageScore))

Append data to regional CNN, predict location

end

end

end

B. CNN-Based Localization

We use 4 different CNNs, one for each region, and a general neural network for localization of data that either lacks lidar images or provides corrupt images that are not classifiable. The regional CNNs are trained using data points specific to the region, whereas the general CNN is trained using all of the training samples. The input to the CNN is the matrix shown in eq. 1, which contains the ray-tracing data

associated with the data point: RSS, elevation AoA, azimuth AoA, elevation AoD, azimuth AoD, ToA and ray phase. The CNN outputs 3 values, which correspond to the x, y and z coordinates of the predicted location. We use CNNs with one convolutional layer which consists of 4 filters with 2x2 kernel size, ReLu activation and 2x2 max pooling. The architecture of the CNNs and the associated learning parameters are summarized in Table I. Most of the utilized CNNs have a fairly similar architecture, with the main difference being the samples used for training the networks. Regions with a smaller area, i.e less vehicles, utilize dropouts to prevent over-fitting. To assess the performance of the network architecture, we plotted the model's training vs. validation loss over the number of iterations. The aim is to ensure the intersection of the training and validation loss curves, which indicates a good fit. A sample plot of the region 2 CNN model is shown in Fig.4. We also use the Sigmoid activation function for the hidden layers, linear activation function for the output layer and MSE for the loss metric.



Fig. 4. Training vs. Validation Loss for Region 2 CNN Model

TABLE I

TABLE SUMMARIZING THE NETWORK ARCHITECTURE OF THE CNNs.

Parameters	Values			
	Region 0	Region 1	Region 2	General CNN
Architecture	128x64x32x16x3	128x64x32x16x3	128x16x3	128x64x32x16x3
Epochs	110	100	50	300
Batch Size	10	10	10	10
Dropout	0.2	N/A	0.2	N/A

V. PERFORMANCE EVALUATION

As mentioned before, we use the Raymobtime-s008 and s009 data sets that are available at [14]. In this section, we refer to the s008 and s009 data sets as dataset-1 (DS1) and dataset-2 (DS2), respectively. We first pre-process the data as described in Section III and then apply the steps in the proposed algorithm. To determine the performance, we

assess various adaptations of the algorithm. We also create a network comprised of a single CNN using our input matrix of ray-tracing data described in eq.1 to use as a benchmark to compare our results with existing studies that deep neural networks, which showcases the improvement achieved as a result of the LIDAR data. On average, the training time for the regions 0, 1, 2 and the general CNN models is 2.48 minutes, 4.11 minutes, 8.61 seconds and 8.44 minutes and the prediction time per data point is 1.91 ms, 64.73 μ s, 350.28 μ s, 119.42 μ s, respectively.

We showcase the results of the proposed technique given the two main approaches for regional classification described in section IV: regions with overlapping vs. solid boundaries and hybrid regional classification vs. image classification. We use root mean squared error (RMSE) and mean absolute error (MAE), shown in eq. 2 and eq. 3, as metrics to assess the performance of our algorithm. Both metrics describe the average model prediction error, however, the RMSE assigns comparatively higher weights to large errors. As such, MAE can be used to determine the overall performance for a given data set, whereas RMSE can be used to detect the instances of outliers and false classifications. In this case, as we take a regional CNN approach using image classification, a higher RMSE can be linked to false classification as the regional CNNs are only trained to make predictions within their boundaries. Hence, data points that are classified into inaccurate regions can cause large errors.

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y - x)^2}{n}} \quad (2)$$

$$MAE = \frac{\sum_{i=1}^n |y - x|}{n} \quad (3)$$

The results of the different combinations of techniques are summarized in Tables II and III. The results display the error for each region in the map. The first noticeable outcome is that the localization error in region 0 is significantly higher in data set-1 than data set-2 (see Table III). The classification accuracy of data set-1 confirms that this is due to the higher number of false positive image classifications in that region (see Table II). In this paper, the classification accuracy is defined as the percentage of true positive classifications. The quality of the images causing the false classifications is substandard in comparison to the training images shown in Fig.3, which makes them unidentifiable to the image differencing algorithm. The classification algorithm described in section IV requires the image differencing score to be higher than the defined threshold in order for the data point to be sorted into a regional CNN. The threshold is based on trial and error computations that achieved the greatest overall localization error; in this study we require a 70% image match. This allows some variation between the vehicle's lidar images and the training images, while minimizing the number of wrongful classifications. As such, the lower quality images coincidentally passed the threshold and were misclassified into region 0. As seen in Table III, for region 1 and 2, the proposed

approach achieved a higher accuracy, and the localization error of data set-1 using the proposed technique is noticeably lower than the localization error using only one CNN. Moreover, the variation in localization error between the two data sets in this region shows the potential of this algorithm as more image data sets become available. With a higher classification accuracy, data set 2 achieved an RMSE and MAE of 2.45 m and 3.60 m in region 0, in comparison to the RMSE and MAE of 10.9 m and 10.38 m in data set-1. In this study, we explored the use of street and building silhouettes to classify the regions as the lidar images shown in Fig.3 which lacked a great amount of detail. However, future works can examine the use of more reliable image classification techniques such as the geographical landmarks used in [9] to reduce the chances of false positives and improve the overall error.

The results show that the regional CNNs with the hybrid regional classification achieve the best localization accuracy with a MAE of 7.41 m and 6.55 m for data sets 1 and 2, whereas the network architecture with only 1 CNN has the highest error. This implies that the regional CNN approach allows the neural networks to better understand the trends and outcomes associated with the specific environments, hence producing more accurate predictions. The results also highlight the significance of the regional boundaries and classification techniques. The use of overlapping boundaries at neighbouring regions increased the classification accuracy from 98.14% to 99.95%, which consequently improved the localization error. This can also be seen in the localization error of data set-1 in region 0 as the use of overlapping boundaries improved the error of the region from 12.34 m to 10.9 m. However, the addition of RSS based fingerprinting to the classification technique achieved negligible improvements to the localization accuracy. This is indicated in the results as the RMSE of the regional CNNs with overlapping boundaries and image classification is 7.84 m, a drawback of only 0.05 m in comparison to the CNN with overlapping boundaries and hybrid classification. A similar trend can be seen in the MAE as well as the results of CNNs with solid boundaries shown in Table III. Although the enhancement of the algorithm in terms of localization accuracy by using hybrid classification is insignificant, one added benefit that remains is the improvement with respect to computational complexity. As described in section IV, the fingerprinting based classification is a simple comparison operation with a predefined threshold, while the image classification method requires the implementation of image pre-processing and image differencing as described in Fig. 2.

The proposed technique also achieved negligible delay, considering pre-processing and prediction time. The maximum recorded time for localization is 329.91 ms, with 328 ms for regional classification and 1.91 ms for CNN-based localization. The recorded latency is based on the maximum time it takes both the regional classification and CNN algorithms to make a prediction. Localization systems typically require delay within the milli-second time frame in order to be feasible in a real-time localization scenario [16], which makes the delay of

TABLE II
TABLE SUMMARIZING CLASSIFICATION ACCURACY.

	Accuracy	
	DS1	DS2
Hybrid Regional Classification with solid boundaries	97.12%	98.14%
Image Classification with solid boundaries	97.1%	97.9%
Hybrid Regional Classification with overlapping boundaries	99.5%	99.95%
Image Classification with overlapping boundaries	98.12%	98.8%

TABLE III
TABLE SUMMARIZING LOCALIZATION ERROR.

Dataset	Region	RMSE (m)		MAE (m)	
		DS1	DS2	DS1	DS2
One CNN		9.44	12.51	10.91	9.82
Regional CNN with Overlapping Boundaries (Hybrid Classification)	0	10.90	2.45	10.38	3.60
	1	8.34	7.83	8.03	8.32
	2	3.02	4.52	4.17	6.22
	Gen. CNN	8.28	6.51	8.56	6.54
	Total	7.79	7.08	7.41	6.55
Regional CNN with Overlapping Boundaries (Image Classification)	0	10.78	3.07	10.46	4.33
	1	8.11	7.52	7.71	8.34
	2	2.97	3.02	4.23	4.08
	Gen. CNN	8.14	7.31	8.43	5.21
	Total	7.84	7.79	7.88	6.59
Regional CNN with Solid Boundaries (Hybrid Classification)	0	12.75	3.64	12.27	5.31
	1	7.68	7.63	7.50	7.75
	2	3.403	4.24	4.81	6.15
	Gen. CNN	9.09	8.59	8.81	5.48
	Total	9.36	8.08	8.43	6.05
Regional CNN with Solid Boundaries (Image Classification)	0	12.34	3.85	12.45	5.80
	1	8.33	7.29	8.42	8.17
	2	3.42	4.09	4.82	5.75
	Gen. CNN	8.29	9.16	7.77	6.52
	Total	8.65	8.26	8.12	7.23

the proposed technique acceptable. The proposed technique is also beneficial in terms of availability and costs as it takes advantage of current 5G infrastructure as well as LIDAR technology, which most vehicles come fully equipped with. By utilizing radio parameters that are readily available through the formation of a communication link between the base station and the user, the number of transmissions required for localization is reduced. Moreover, the use of machine learning eliminates the need for complex mathematical computations typically used in conventional methods such as trilateration and triangulation. As our proposed methodology uses CNNs,

the training process is completed offline and is only required to be completed once. This optimizes the energy efficiency of the system. Additionally, as we propose a centralized technique that takes place at the base station rather than the user's devices, impact to the user's battery life is minimized.

The performance of the proposed methodology is comparable to traditional and recent outdoor localization methods. We compare the performance of our technique to other vision based and deep learning based outdoor studies. We use error per square meter as a metric to measure the localization accuracy given different terrain sizes. Our technique achieves the best error per square meter using the cascaded regional classification and CNN-based localization approach in comparison to the work in [13], [17], [18], and [9], where fingerprint-based localization using FFNNs, NN-based localization using MIMO, supervised machine learning-based localization using MIMO and R-CNN vision-based localization methods are proposed. Those techniques achieve a mean error of 19.9 m, 9.60 m, 5 m and 28.4739 m and an error per square meters of 0.00028348, 0.00016165, 0.00005007 and 0.013559, respectively. Our proposed methodology obtains a mean error of 6.05 m and an error per square meter of 0.00002241. Although [18] achieves a smaller localization error of approximately 5 m, their localization area is considerably smaller than our area which in turn can provide smaller errors. [9] implements a localization approach using vision data only. Their results show that relying on only vision data can yield higher errors. Our cascaded approach takes advantage of both vision data and radio parameters which allows for more accurate representation of the environment, hence providing better results.

VI. CONCLUSION

In this paper, we introduced a localization technique using a two-stage solution comprised of regional classification and CNNs using LIDAR and RSS data. We propose two interpretations for regional classification: a hybrid approach using fingerprinting and image differencing and standalone image differencing. We also explore two ways of defining regional boundaries: regions with solid boundaries and regions with overlapping boundaries. Our algorithm is applied to the Raymobtime s008 and s009 simulated datasets [14], that model the neighbourhood of Rosslyn in Arlington, Virginia. We split the map of Rosslyn into 3 distinct regions that are distinguishable through their landmarks and use the lidar images collected by the vehicles to detect the landmarks in the localization territory to predict the receiver's region. Once the vehicle is classified into a region, the user's data is used as an input into the regional CNN to predict their location co-ordinates. We create an input matrix for the CNNs that contains the user's RSS, AoA, AoD, ToA and ray phase for all 25 antennas in the vehicle. Our results show that the regional CNN approach with hybrid regional classification and overlapping boundaries achieved the highest classification accuracy and lowest localization error, with a MAE of 7.41 m and 6.55 m for data sets 1 and 2, respectively.

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